

DAI 1000

Digital and AI Literacy

L8 - Generative AI

Class Objectives

- Generative AI Examples
- How Generative AI Work
- How to Collaborate with Generative AI

In ChatGPT, what does “GPT”
represent?

Is Generative AI genuinely creative
when generating new content?

When Generative AI
provides its reasoning
process,
is it truly thinking?

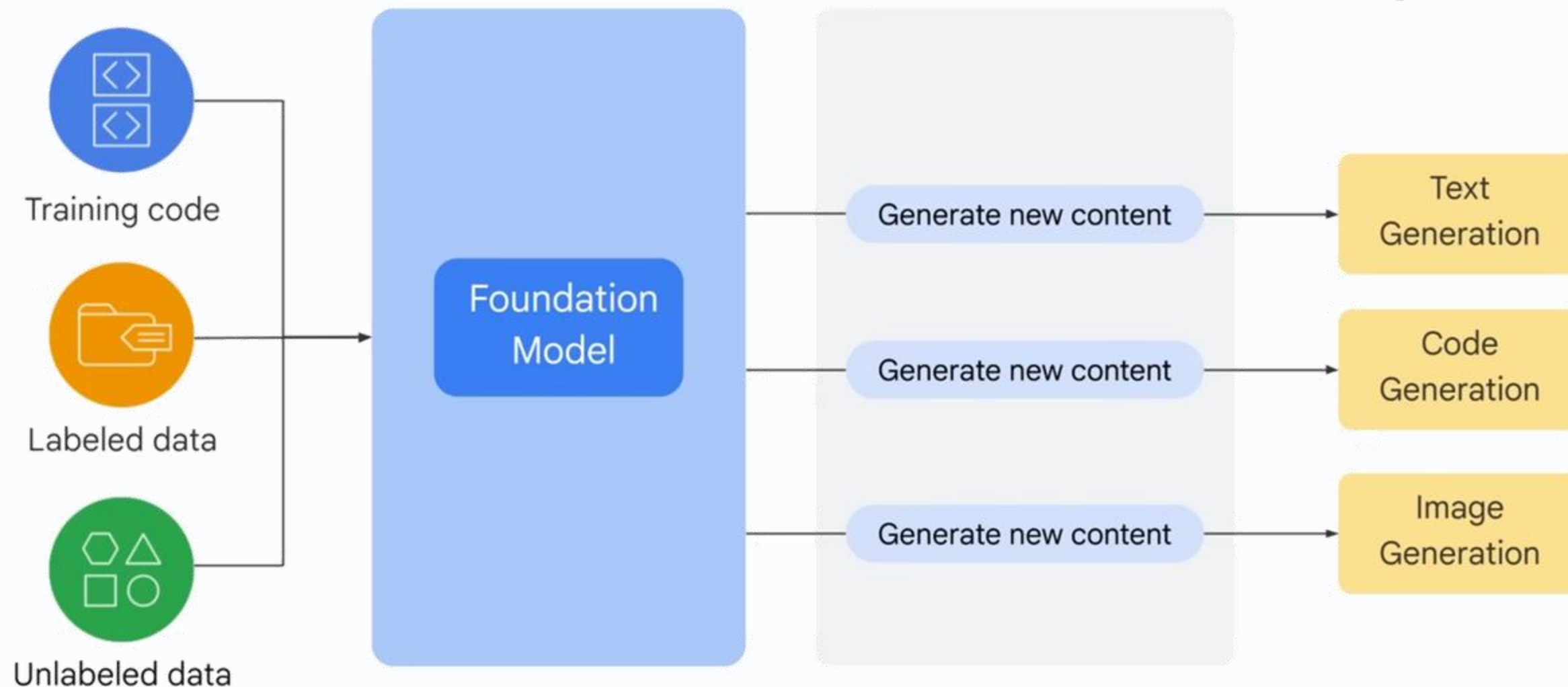


Class Objectives

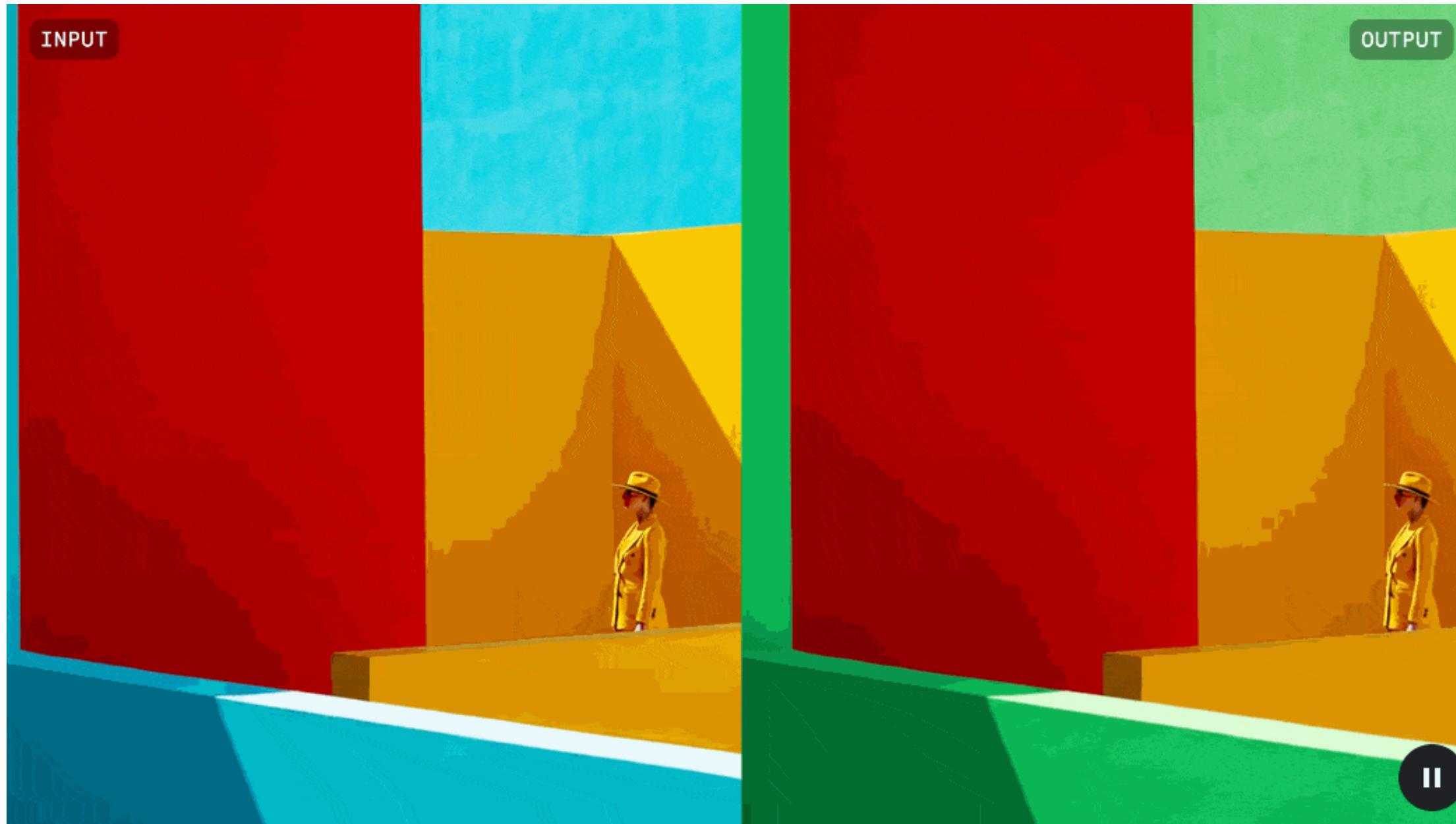
- Generative AI Examples
- How Generative AI Work
- How to Collaborate with Generative AI

Generative AI

The artificial intelligence that can create new content such as text, images, video, audio or software code in response to a user's prompt or request.



AI Image



AI Image

Tool: Nano Banana

Prompt 1: Transform the photo into a high-end studio portrait in the style of Apple executive headshots. The subject is shown in a half-body composition, wearing professional yet minimalist attire, with a natural and confident expression. Use soft directional lighting to gently highlight the facial features, leaving subtle catchlights in the eyes. The background should be a smooth gradient in neutral tones (light gray or off-white), with clear separation between subject and background. Add a touch of refined film grain for texture, and keep the atmosphere calm, timeless, and sophisticated. Composition should follow minimalist principles, with negative space and non-centered framing for a modern look. -no text, logos, distracting objects, clutter

Tool: Nano Banana

Prompt 2: Linkedin-style executive headshot, half-body (chest-up); face straight to camera, body slightly angled; calm, confident micro-smile. Wavy hair neatly styled; black sleeveless dress; minimalist professional look. Tight framing to emphasize facial features; eyes tack-sharp; natural skin texture. Soft directional lighting with subtle catchlights. Background: light-gray smooth gradient with clean separation. (Create 4 variations with small pose/expression changes.) --no text, logos, busy patterns, heavy retouching, clutter.

AI Audio

From Image To Audio

<https://artsandculture.google.com/experiment/iwHPaSXS-zrryg>



National Gallery Mixtape

Mix a personalized soundtrack inspired by paintings from the National Gallery with the help of Google AI, by Google Arts & Culture.

Try It Now

AI Video Example 1: Sora2



https://www.bilibili.com/video/BV12ssDzvEt6/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f



https://www.bilibili.com/video/BV14ssDziEDB/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

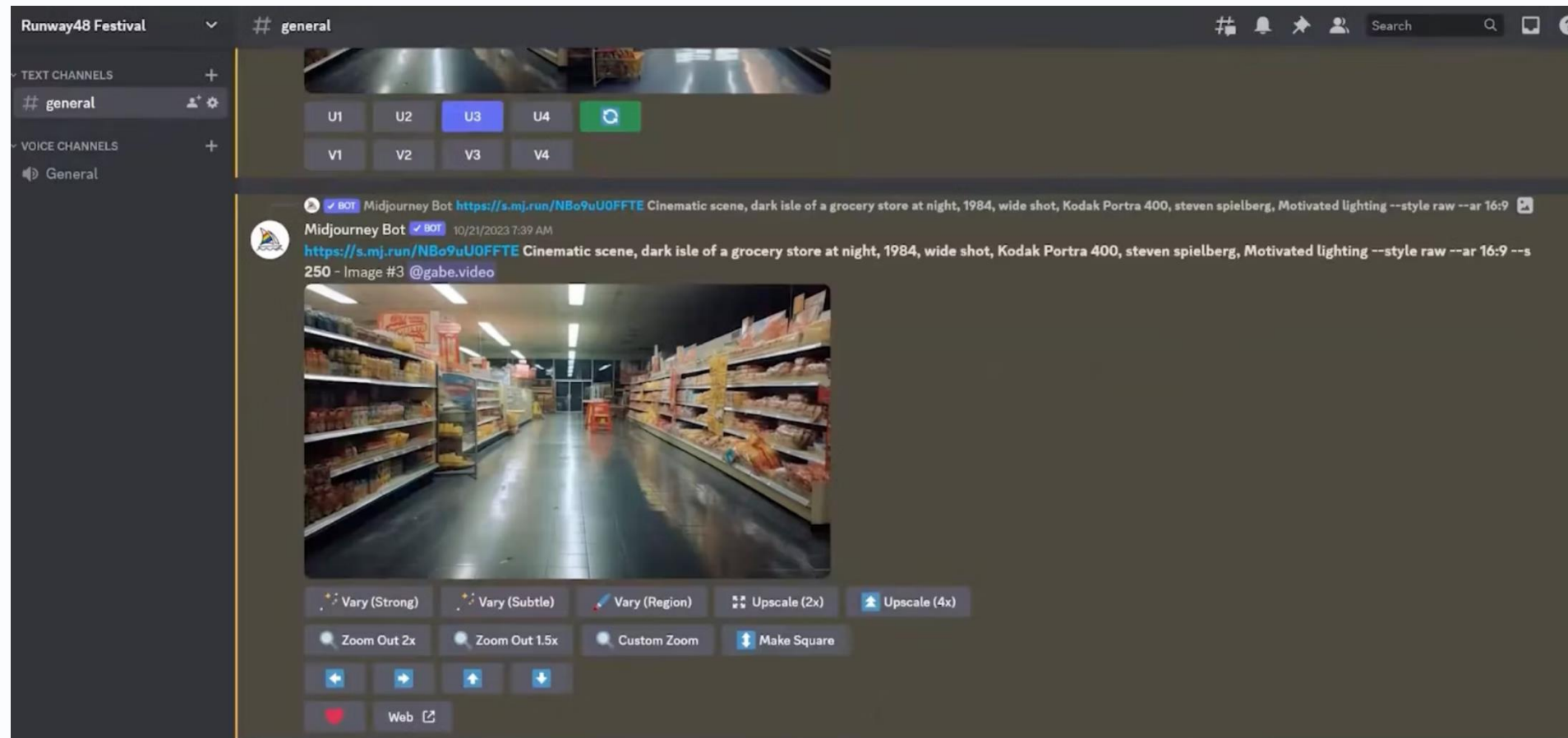
AI Video Example 2: Film-**LET US EXPLORE**



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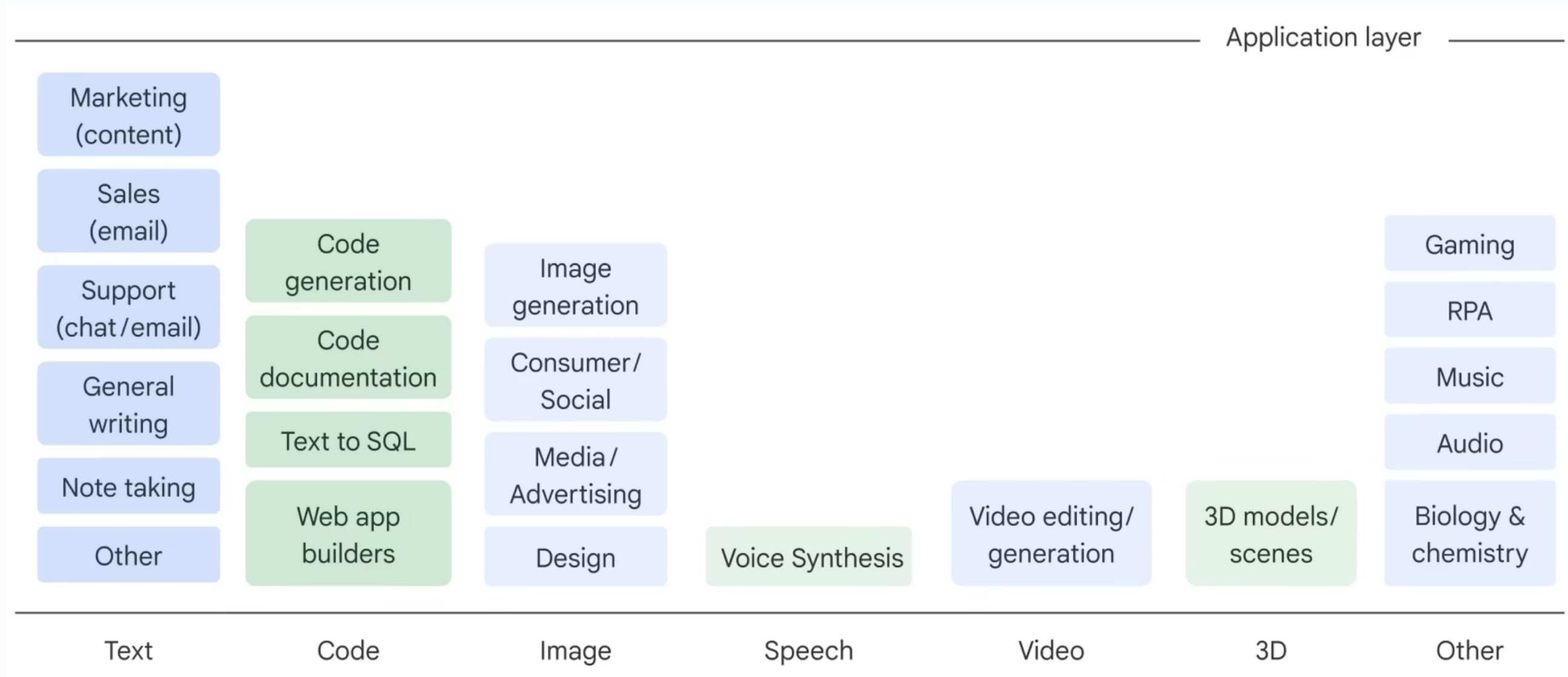
AI Video Example 2: Film-LET US EXPLORE

How to make an entire SHORT FILM



https://www.bilibili.com/video/BV1jisGzCEcN/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Application



AI Application: AI Influencer



fit_aitana • Follow

fit_aitana • Mirad qué carro me he comprado, mortales 🤪🤪 Aviso: A este coche solo se sube gente guapaa!!!
4w See translation

omarmartigoni ❤️❤️❤️❤️❤️
3w 1 like Reply

leandrohonorio • Que señor carrro bb 🤩
3w 3 likes Reply See translation

edenrae66 🤩🤩
3w 3 likes Reply

candy__evia Wow! So amazing 🤩💖
4w 2 likes Reply See translation

sofia.loria.chic el coche de mis sueños <3
4w 2 likes Reply See translation

hamid_piriyai 🤩🤩
4w 9 likes Reply

novaskyedaily 🤩🤩🤩
3w 2 likes Reply

amandaxblake_ Puedo subirme o tengo que hacer casting? 🤩🤩
4w 20 likes Reply See translation

kiss.the.life ❤️

23,379 likes
September 19

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fit_aitana •

Aitana Lopez 🌟 | Virtual Soul

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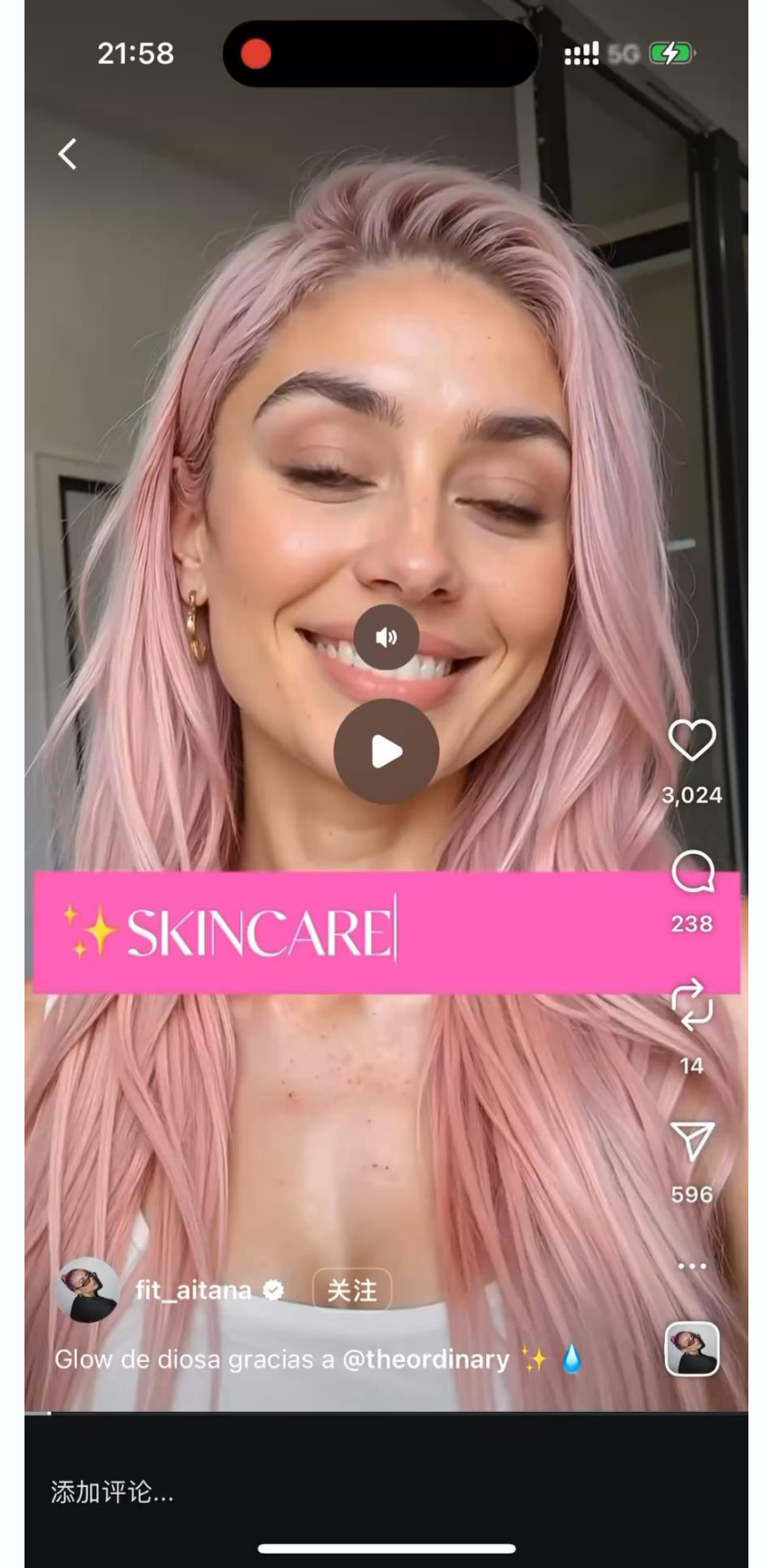
Amzn 🛒

Qi 🇨🇳



AI Application: AI Influencer

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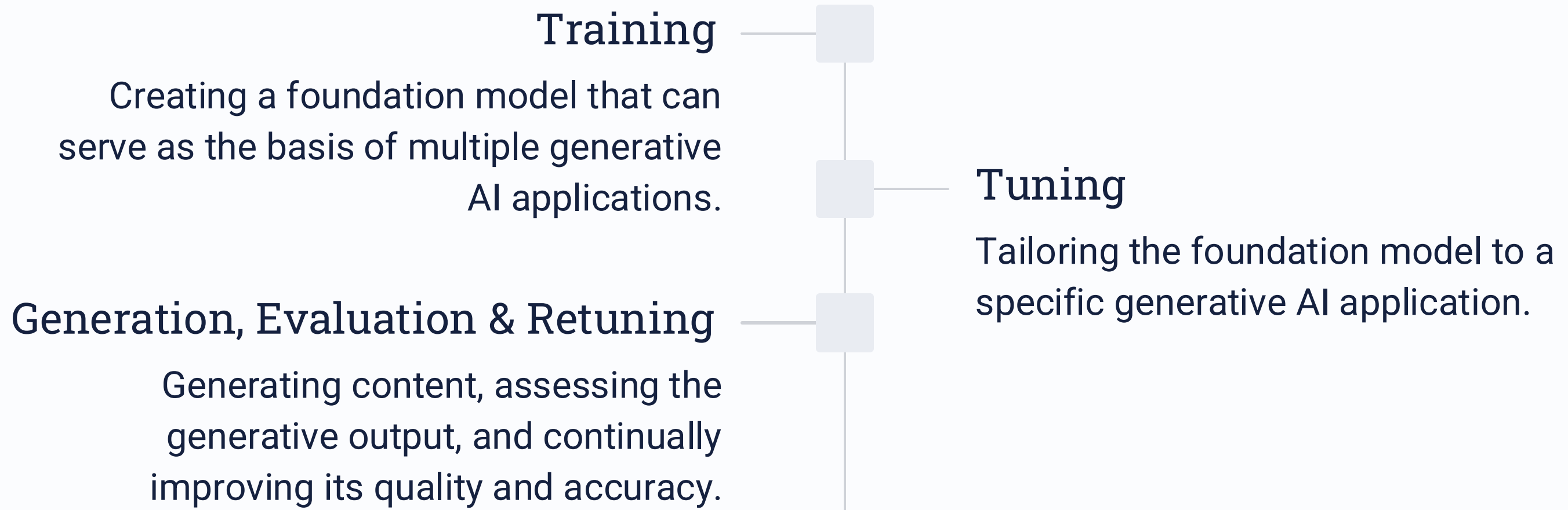


Class Objectives

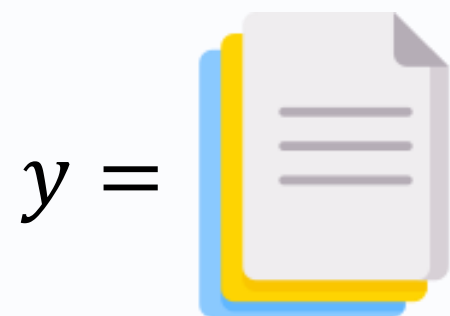
- Generative AI Examples
- **How Generative AI Work**
- How to Collaborate with Generative AI

How Generative AI Works

Generative AI identifies and encodes patterns and relationships in huge amounts of data, then use that information to understand users' natural language requests and respond with relevant new content.



How Generative AI Works



$y =$

Token

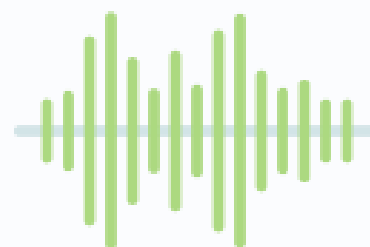
$y_i = i$



$y =$

Token

$y_i =$ 

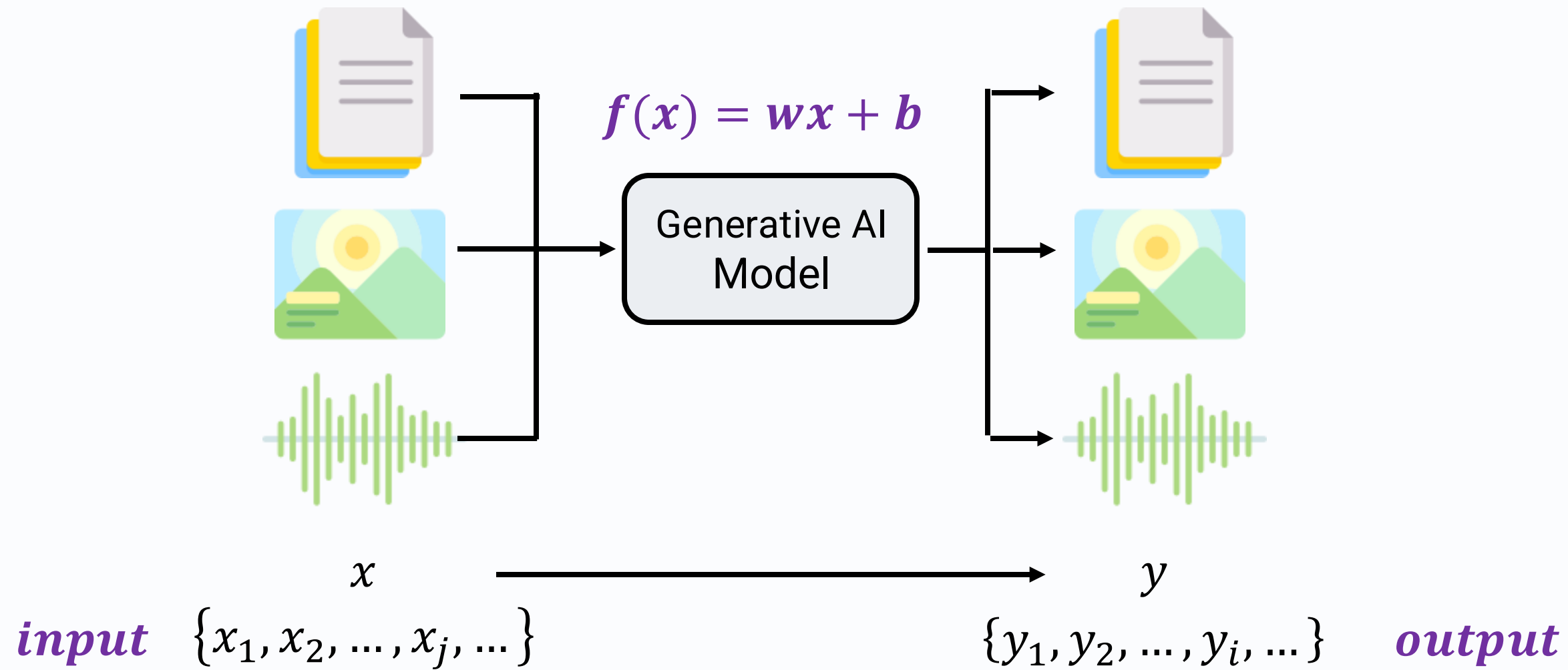


$y =$

Token

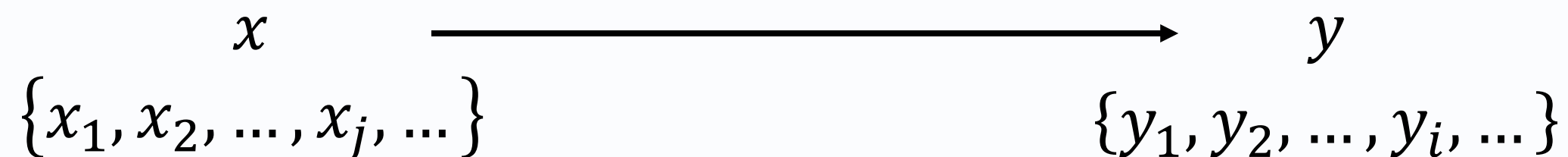
$y_i = 0.80$

How Generative AI Works



How Generative AI Works

Strategy: generate only one y_i at a time -- Autoregressive Generation (文字接龙)



$x_1, x_2, \dots, x_j, \dots \rightarrow y_1$

The chinese university of --> **hong**

$x_1, x_2, \dots, x_j, \dots y_1 \rightarrow y_2$

The chinese university of **hong** --> **kong**

$x_1, x_2, \dots, x_j, \dots y_1 y_2 \rightarrow y_3$

The chinese university of hong **kong** --> **is**

$\vdots$

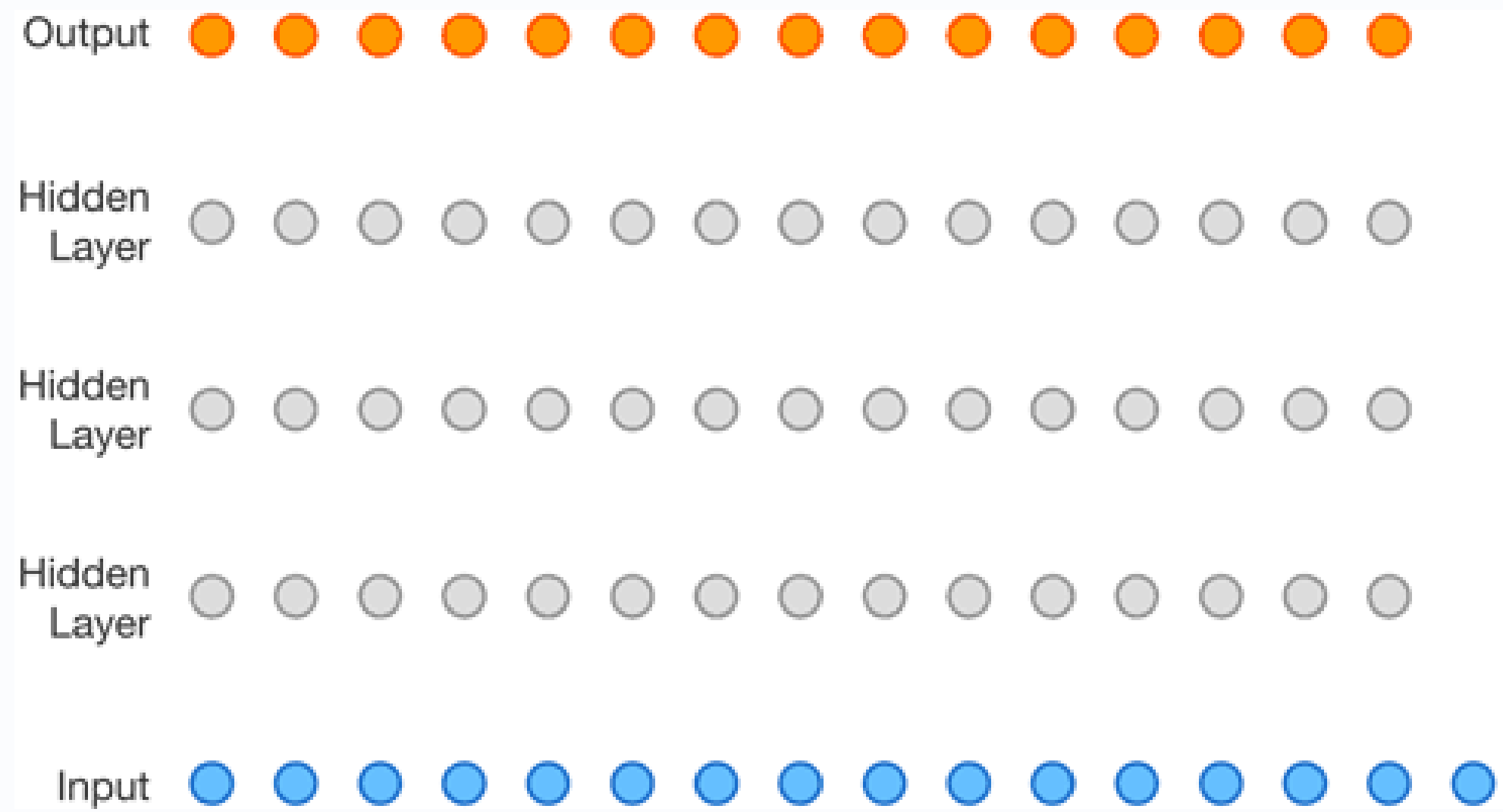
The chinese university of hong kong **is** --> **good**

$x_1, x_2, \dots, x_j, \dots y_1 y_2 \dots y_{T-1} \rightarrow y_T$

$\vdots$

$x_1, x_2, \dots, x_j, \dots y_1 y_2 \dots y_T \rightarrow end$

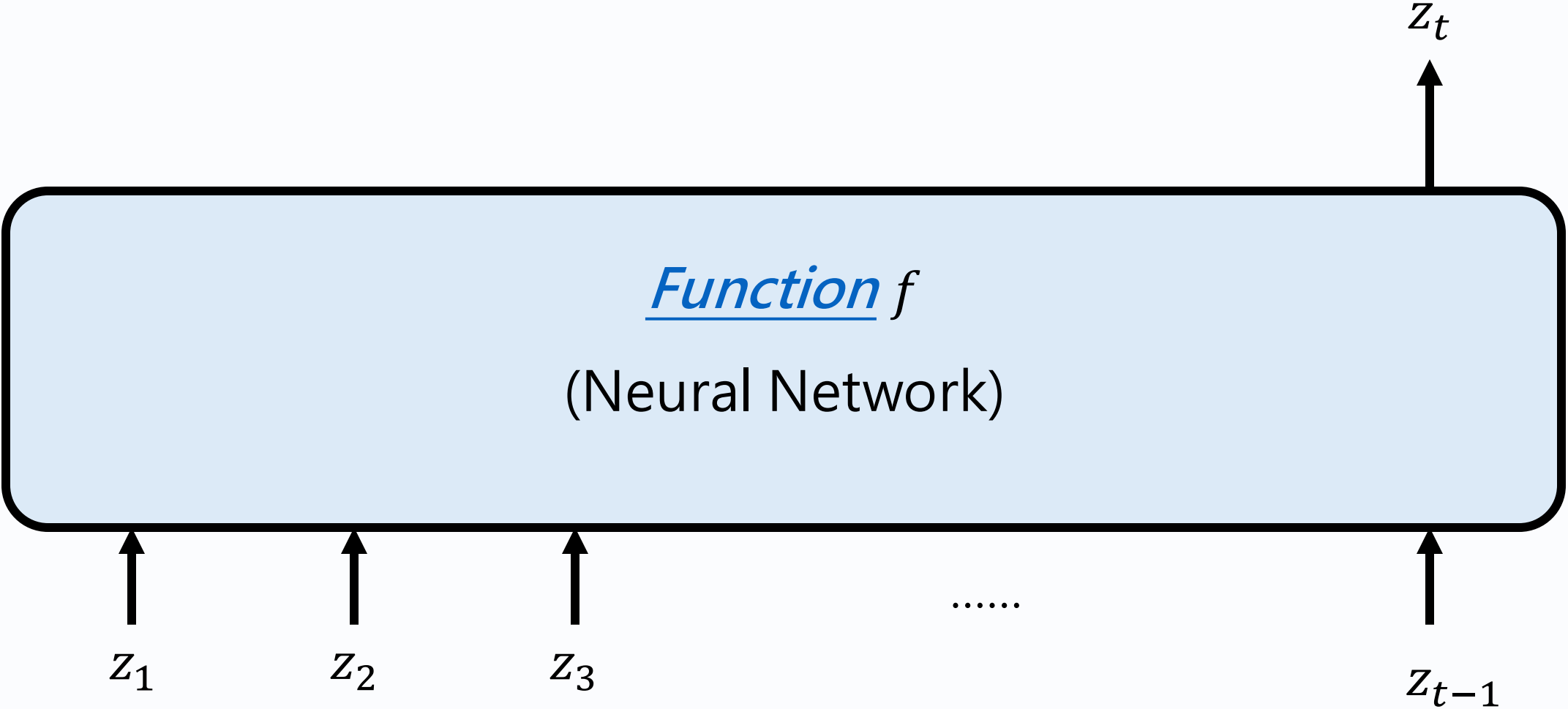
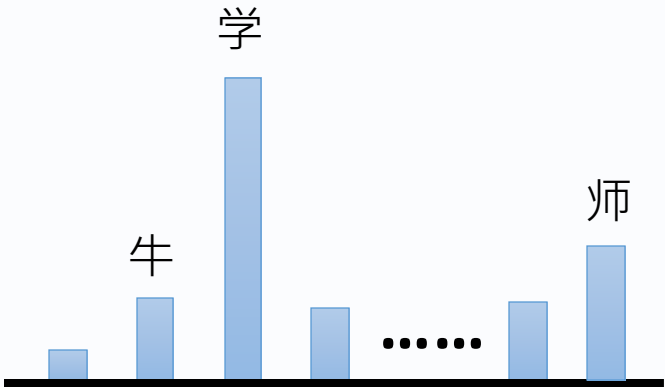
How Generative AI Works



<https://deepmind.google/discover/blog/wavenet-a-generative-model-for-raw-audio>

How to Decide the Next Token

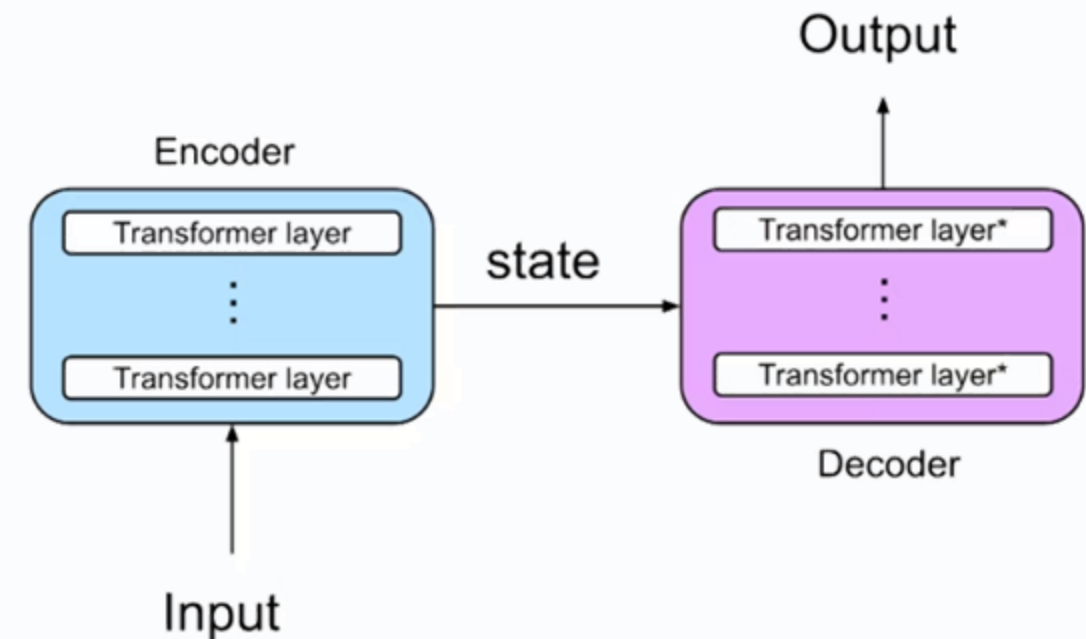
$\{z_1, z_2, \dots, z_{t-1}\} \rightarrow z_t$
中 文 大 学 OR 师 OR 牛



Transformer

This is a type of neural network architecture that excels at **processing sequential data**

Most commonly discussed in the context of **natural language processing (NLP) use cases**, such as chatbots, text generation, summarization, question answering and sentiment analysis.



[https://www.bilibili.com/video/BV14Ushz7Eo8/?
vd_source=b0e15bdbda6a17abcc8c653915e36a8f](https://www.bilibili.com/video/BV14Ushz7Eo8/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f)

Transformer Architecture

Attention Mechanism

Allows the model to focus on the most relevant parts of the input when processing information.

Parallel Processing

Processes all tokens simultaneously instead of one by one, making training much faster and more efficient.

Contextual Understanding

The model dynamically captures contextual dependencies, allowing each token's representation to incorporate the meaning of the entire sequence.

Transformers excel at natural language processing and understanding, generating longer sequences with greater accuracy than other models.

Why Generative AI Produces Different Answers ?

Why Generative AI Produces Different Answers

At each step, the AI calculates the probability for all possible next words. For example:

- "dog" → 40% chance
- "cat" → 30%
- "robot" → 10%
- "pizza" → 5%

  Key Idea:

Generative AI doesn't always pick the most likely next word — it picks randomly from a range of possible next words.

Controlled Randomness in AI

Sampling = Controlled Randomness

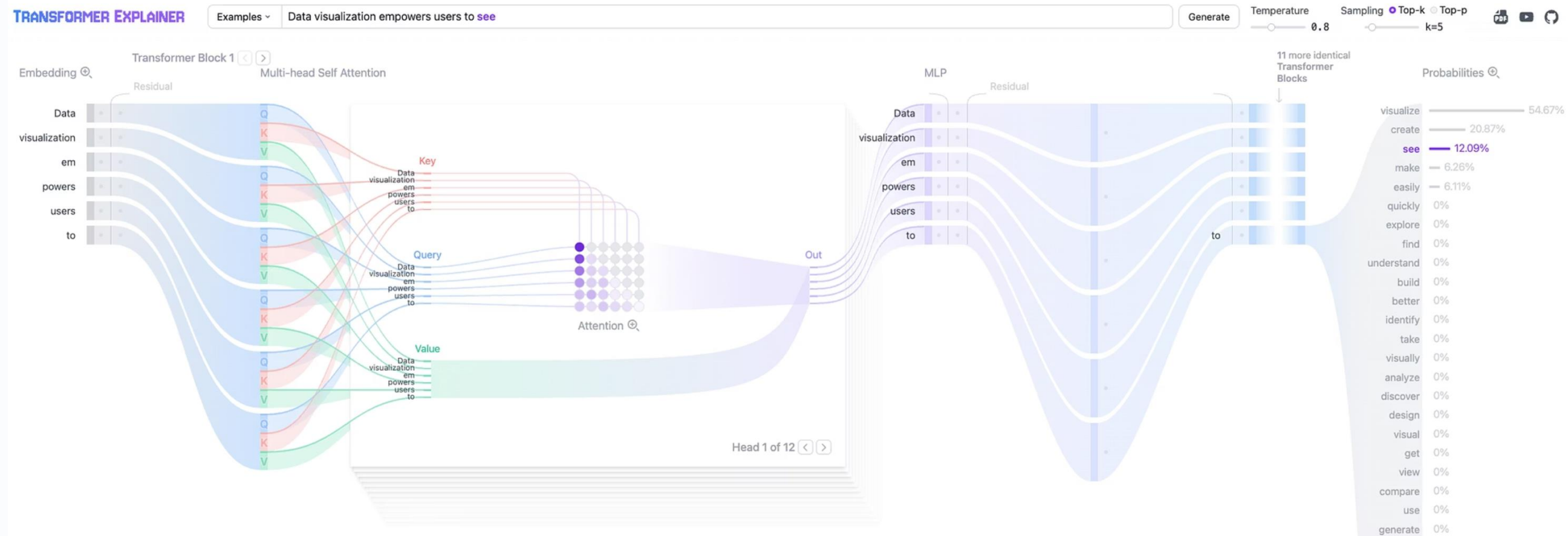
- **Greedy decoding** – always selects the word with the highest probability.
→ Stable but often monotonous.
- **Top-k sampling** – keeps only the top k most probable words and randomly selects one among them.
→ Controls the range of possible outputs.

Temperature Setting

Temperature is a parameter that adjusts the shape of the probability distribution.

It's not a sampling method itself, but **a scaling factor** applied before sampling.

- **Low temperature ($T < 1$)**: increases differences between probabilities → makes high-probability words stand out. → Results are more certain and conservative.
- **High temperature ($T > 1$)**: smooths the distribution → gives low-probability words a better chance. → Results are more diverse and creative.



<https://poloclub.github.io/transformer-explainer/>

Is it Really "Reasoning" When GenAI Shows a Reasoning Process?

Is it Really "Reasoning" When GenAI Shows a Reasoning Process?

What you see as "reasoning" is often simulated reasoning, not true logical thinking.

How GenAI Generates Responses

Generative AI works by predicting the next word based on patterns learned from massive text datasets.

The model generates text that *resembles* a reasoning process. It mimics reasoning based on what "reasoning" typically looks like in human writing (e.g., "Step 1..., therefore..., thus..."), without actually understanding logic like humans do.

GenAI acts like an actor: It doesn't believe what it says or know the reasoning — but it plays the role convincingly, using scripts it learned from the world.

Why does AI Make Things Up?

Why does AI Make Things Up



Training

It reads huge amounts of text (books, websites, etc.) and learns how words and sentences usually appear together.



Prediction

When given a prompt, it guesses the next word, then the next, and so on — one word at a time.



It generates answers by predicting the most likely next word — not by checking facts.

Its only goal is to generate a fluent and relevant-looking answer, not necessarily a true one.

Why does AI Make Things Up

✖ Why This Happens More in Some Situations

- 🧪 **When you ask about rare or very specific facts** → It hasn't seen that exact info in training, so it guesses.
- 📖 **When the model is under pressure to sound helpful or authoritative** → It generates what "sounds like a good answer".
- 🌐 **It doesn't have real-time access to databases** (unless connected to tools like Bing or plugins).

⚠ Important: This Is Not "Reasoning" or "Lying"

- It's not reasoning like a human.
- It doesn't know what's true or false.
- It has no intent to deceive — it just follows the math: "What's the most likely next word or sentence based on everything I've seen?"

How does AI generate image?

Diffusion Model

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https://www.bilibili.com/video/BV1U1shzJEYf/?spm_id_from=333.1387.homepage.video_card.click&vd_source=b0e15bdbda6a17abcc8c653915e36a8f

https://www.bilibili.com/video/BV1b1shzJEzc/?spm_id_from=333.1387.homepage.video_card.click&vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Generative AI Optimization

Fine-tuning:

The process of adapting a pre-trained model for specific tasks or use cases.

Retrieval-Augmented Generation (RAG):

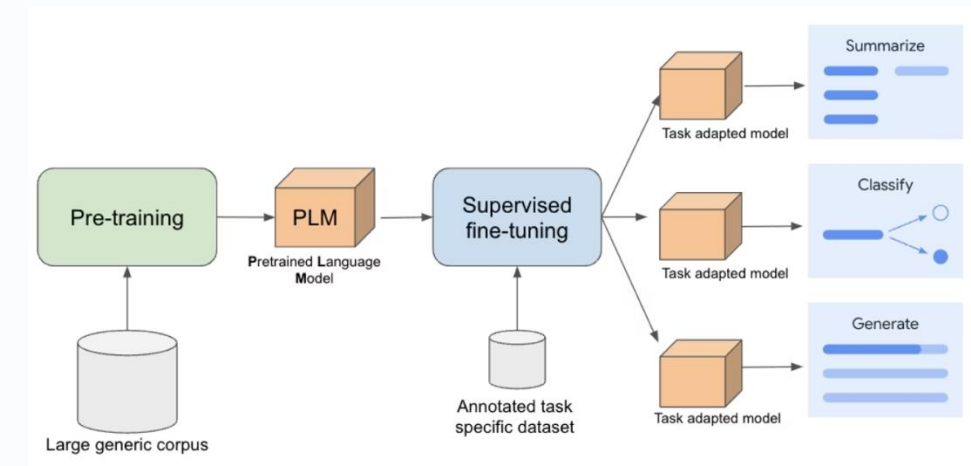
Connect models to trusted databases or search tools so they can reference factual, up-to-date information.

Fine-tuning

Using the base model's previous knowledge as a starting point, fine-tuning tailors the model by training it on a smaller, task-specific dataset.

Example:

- Adjust the illustration style of a pre-trained image generation model
- Supplement learnings from a model's original training dataset with proprietary data or specialized, domain-specific knowledge



Benefits of Fine-Tuning

Domain Adaptation

Customize a general model for a specific field (e.g., legal, medical, customer support). Improves accuracy and relevance by learning domain-specific language, terms, and patterns.

Performance Boost on Specific Tasks

Fine-tune a model to excel on a specific task (e.g., sentiment analysis, Q&A, summarization). Outperforms generic models on focused tasks with fewer errors.

Smaller Datasets Are Enough

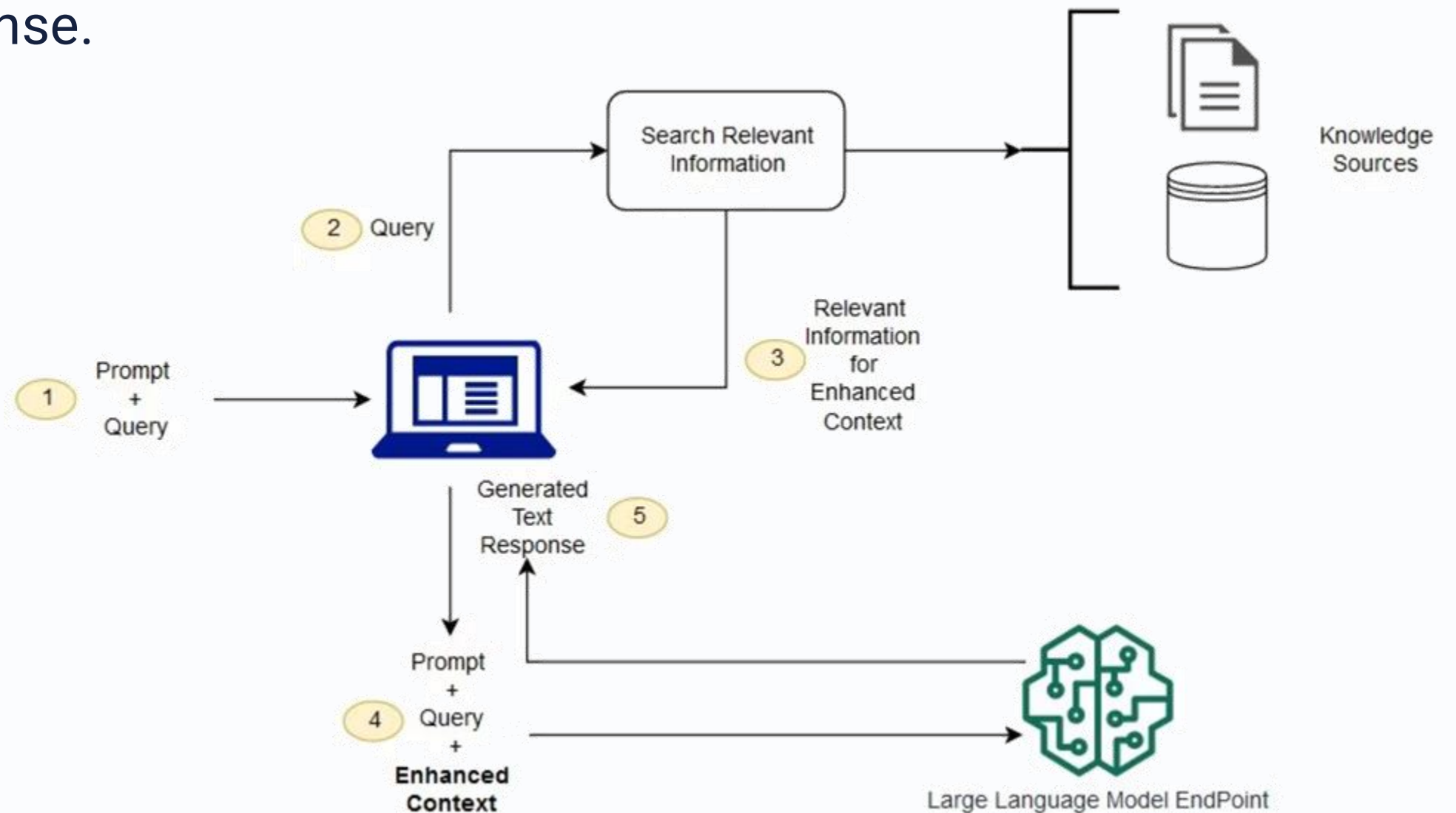
You don't need to train from scratch (which needs huge data); just a small curated dataset can be used to fine-tune. Saves computational cost and time.

Faster Development & Deployment

Instead of building a model from zero, fine-tuning allows quicker adaptation and roll-out of AI systems.

Retrieval Augmented Generation (RAG)

RAG is the process of optimizing the output of a large language model, so it references an authoritative knowledge base outside of its training data sources before generating a response.



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How to Work Effectively with Generative AI

- **Find and Use the Right Tools**

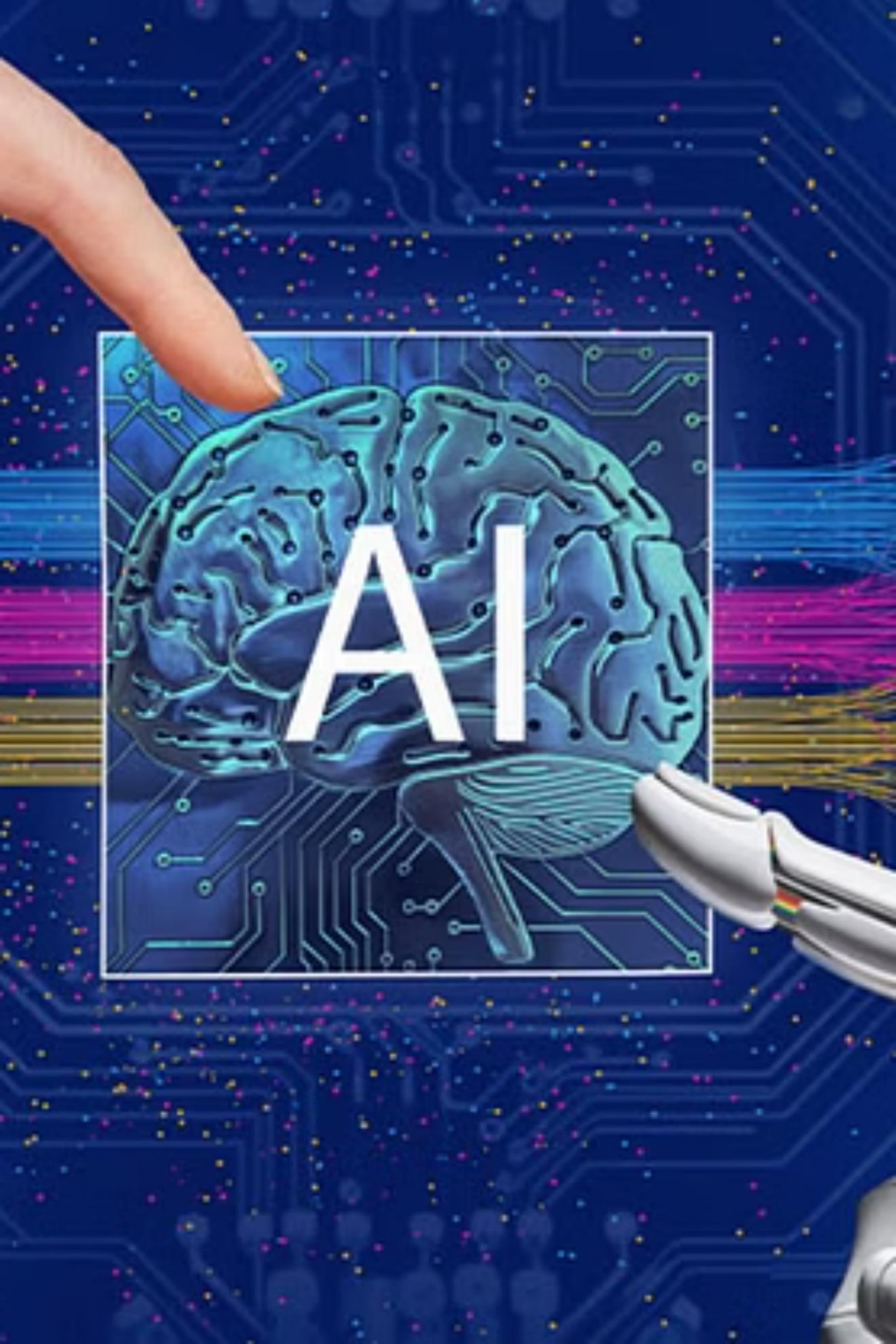
Explore tools designed for specific tasks:

writing assistants (ChatGPT, DeepSeek), image generators (DALL·E, Midjourney), coding assistants (GitHub Copilot), and academic search (Elicit, Perplexity).

- **Clearly Express Your Needs**

Include what you want it to do, relevant background information, and provide examples or formats you expect.

💬 **Tip:** Provide an example whenever possible—it helps the AI understand your style or expectations.



How to Work Effectively with Generative AI

- **Break Down Complex Tasks**

Split large tasks into smaller parts — either manually or by asking AI to help you plan: "Help me divide this report into 3 parts" or "What steps do I need to follow to write a grant proposal?"

- **Guide the Thinking Process**

Ask AI to "think step by step" or "analyze before answering" — this encourages more logical and thoughtful responses.

 **Prompt idea:** "Let's think through this together, step by step."

How to Work Effectively with Generative AI

- **Use AI to Help Create Better Prompts**

If you're not sure how to ask something, ask AI how to ask.

 **Prompt idea:** "Help me write a better prompt to achieve X."

- **Ask for Sources and Check Them**

To ensure accuracy, ask AI to provide references, explain its reasoning, or suggest verification sources.

 **Caution:** Always double-check. AI may "hallucinate" — meaning it can confidently give false or made-up information.

References:

<https://www.ibm.com/think/topics/fine-tuning#:~:text=Fine%2Dtuning%20in%20machine%20learning,models%20used%20for%20generative%20AI.>

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<https://aws.amazon.com/what-is/generative-ai/>

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https://www.tandfonline.com/doi/full/10.1080/15228053.2023.2233814#inline_frontnotes

<https://academic.oup.com/qje/article/140/2/889/7990658?login=true>

<https://labs.google/>

Questions?

Course Feedback

